Lessons on the Economics of Virtual Transmission

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Abstract

Decarbonizing power systems is expected to require rapid but costly expansion of transmission networks. Virtual transmission instead leverages batteries to more efficiently utilize the existing network, expanding transfer capacity. We describe a general form of virtual transmission in security-constrained economic dispatch and document the performance of two world-first projects in Australia, costing consumers over AUD$750m. Neither project has, to-date, delivered a measurable consumer benefit through the reduction of network congestion. We trace this to ineffective implementation in dispatch. Next, we simulate alternative implementations with a dispatch engine clone and show how, if feasible, better-targeted services could deliver substantial consumer benefits. Realizing this value, however, depends on market design and procurement mechanisms. We highlight that like traditional transmission expansions, virtual transmission can result in benefits (and costs) accruing unevenly across regions and market participants. However, unlike traditional transmission expansion, it is possible to develop market-based compensation mechanisms for virtual transmission service providers.

Keywords: Virtual Transmission, Storage as a Transmission Asset, Transmission Congestion, Regulated Assets, Multi-use Assets, Battery Storage, Clean Energy Transition, Security-Constrained Economic Dispatch

The global energy transition marks a generational shift for power systems in terms of supply mix and grid operation. As countries pursue ambitious carbon reduction targets over the next decade through electrification and the deployment of zero-carbon generation, transmission networks must expand to deliver this capacity to demand centers. The European Commission estimates the EU requires €584b in grid infrastructure investment by 2030[1], while the US Department of Energy estimates the US needs to more than double its transmission network by 2035[2]. To meet existing governments' climate commitments (IEA, 2023)[3], global spending on grids as a proportion of total power system expenditure has to rise from 30% to 40% by 2030. However, the challenges in the deployment of grid infrastructure are substantial, with new infrastructure taking 5-15 years between planning and completion, compared to 1-5 years for renewables[3]. "NIMBYism"[1] adds an additional social hurdle; Germany's 2GW SuedLink HVDC project moved underground at an estimated fivefold cost increase[3], and the US SunZia HVDC project took seventeen years to permit[3]. In Australia, landowners are being compensated AUD$200,000-300,000 per kilometre per year to host transmission lines[3], while major projects face cancellation due to the confounding of these factors[4],[5].

"Virtual transmission" (VT) may be able to address some of these challenges. VT can enhance the efficiency of existing transmission through the utilization of batteries to expand existing network transfer capacity - somewhat analogous to safely converting an emergency shoulder lane to a third lane on a congested, dual-lane highway. In power systems with increasing network congestion, VT is a relatively fast-to-deploy, and potentially, high-value service that complements existing battery use-cases of load-shifting and frequency balancing. Implementation requires development of either a market or regulatory mechanism to compensate batteries for providing VT services. For regulatory mechanisms, benefits to the ratepayers that fund the service will follow from greater network transfer capacity allowing for lower-cost dispatch. Such regulatory mechanisms are particularly important in markets with regional pricing (most non-US markets), where market prices fail to reflect localized congestion costs.

This paper explores VT in an Australian setting, covering two world-first VT projects, the Victorian Big Battery (VBB) and Waratah Super Battery (WSB). We are interested in the best use of these projects and the consumer benefits they deliver, which we present through five lessons.

1) Batteries reserve capacity to provide virtual transmission that raises transmission limits

2) Contracted megawatts of virtual transmission $\neq$ megawatts of transmission expansion

3) Virtual transmission benefits will only accrue at genuine transmission bottlenecks

[1] Not-in-my-backyard.

4) **Market-based compensation and competitive entry are potentially feasible for virtual transmission**

5) **Costs and benefits from virtual transmission accrue unevenly across regions and market participants**

Crucially, we find that neither project has as-yet delivered a measurable consumer benefit through the improvement of network transfer capacity and reduction in wholesale market prices. Our research departs from engineering VT literature[6],[7],[8], including those that examine the economics of merchant VT opportunity cost[9],[10],[11]; focusing instead on consumer benefits. Importantly, our research may be the first to document real dispatch-level implementation of VT (Lesson 1) and quantify actual project congestion-relief benefits (Lessons 2-3). Subsequently, we use NEMPY, a dispatch engine clone by Gorman et al.[12], to undertake counterfactual modeling to identify the potential for improved system benefits. With a focus on ratepayer value, we conclude with a discussion on the suitability of standard regulatory process for transmission in procuring and funding VT (Lessons 4-5).

Australia's National Electricity Market (NEM) is a five-region interconnected market with regional pricing, transitioning from a coal-dominated supply mix, and now with some of the highest rates of solar, wind and battery deployment. Though the system reached a maximum instantaneous share of renewable generation of 78.6% in late 2025[13], an additional five thousand kilometers of transmission are projected over the next decade to support Australia's 62-70% emissions reduction target by 2035[14]. In Australia, transmission is procured regionally, structurally separate from generation and retail, and subject to economic regulation with costs recovered from consumers (the "ratepayer"). Market signals can inform the immediate benefits of marginal transmission expansions, with congestion and network-based renewable curtailment increasing substantially in recent years[15],[14]. However, potential transmission expansion projects are typically evaluated through a standardized regulatory framework that estimates net system benefits. In principle, VT assets can be subject to the same frameworks, with costs recovered via the regulated asset base - similar to the United States' storage as transmission asset (SATA) arrangements[16].

Lesson 1: Batteries reserve capacity to provide virtual transmission that raises transmission limits

Stylized VT setups describe dual-batteries at either end of a set of transmission lines[17]. Some configurations describe the battery physically modulating power through the network via charging/discharging. VT in the NEM is subtler, implemented as a line rating expansion within pre-contingent N-1 thermal constraints[2] and thereby relaxing limits on the corridor's maximum power flows[3]. This is achieved via a single battery holding capacity on standby, monitoring the corridor, and

[2] N-1 describes a single line outage (i.e., minus 1) in a corridor of N lines.

[3] Contingency support for thermal overloading is also identified by Twitchell (2024) as a potential SATA service in US markets[18]

ramping to maintain corridor outflow following an outage. The consumer benefit in system-normal conditions (i.e. in absence of an outage) lies in reduced transmission network redundancy requirements which enable lower energy settlement payments by load - the focus of this research. We are unaware of any prior work that documents the dispatch-level implementation of VT; we begin by introducing a general form here and demonstrate how VT can lead to cheaper market-clearing inside security-constrained economic dispatch (SCED).

Figure 1 is a stylized single-region, three-generator, two-node example of a transmission network with an identical triple circuit line corridor and a set of N-1 thermal constraints limiting line flows. For simplicity, we assume no losses. We step through SCED market clearing with VT, and the NEM's regional price formation at the region's Regional Reference Node (RRN).

An N-1 thermal constraint for overloading on lines x and y, for a loss of z follows:

$$\underbrace{L_x + L_y}_{\text{Monitored Line(s)}} + \underbrace{\delta L_z}_{\text{N-1 Line}} \leq \underbrace{\text{LR}_{x,m} + \text{LR}_{y,m}}_{\text{Maximum Line Ratings, } m\text{-minute duration}} \equiv \mathbf{LR}_{\text{N-1},m}$$

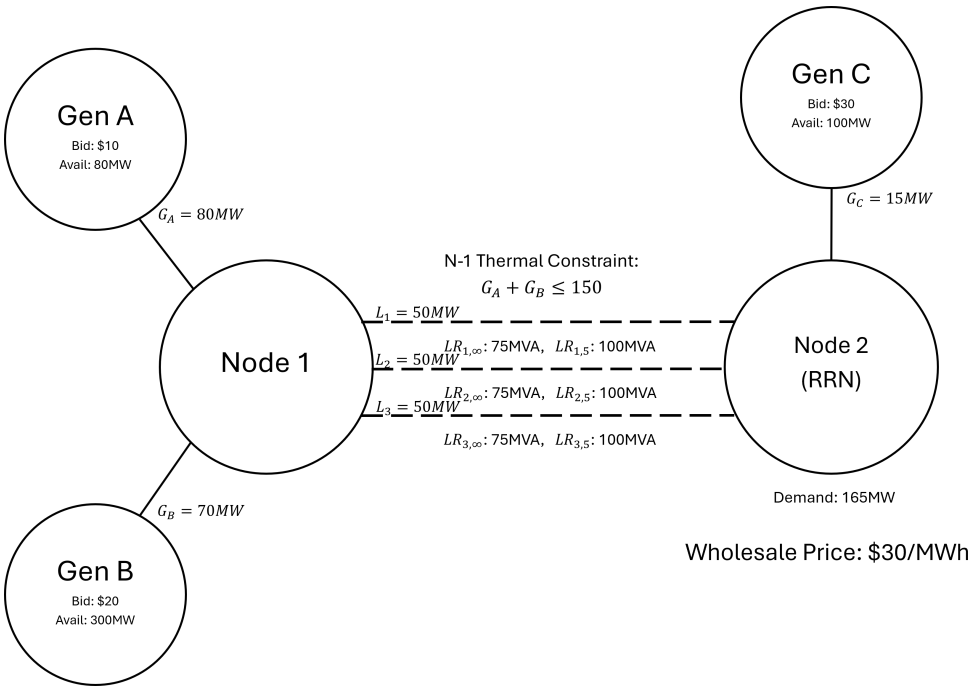
The redistribution factor (δ) follows Kirchhoff's laws in describing the proportion of flows expected to redistribute to monitored lines from an outage of z. We also define $\mathbf{LR}_{\text{N-1},m}$ as the corridor's N-1 limit - the sum of the remaining monitored lines' m-minute line ratings. In SCED, line flows (L) are approximated through a linear combination of generator and interconnector variables, which are collectively allowed to reach $\text{LR}_{\text{N-1},m}$ megawatts for m-minutes after some margin of error (ε). We now group dispatch terms on the left-hand-side (LHS), fixed values on the right-hand-side (RHS):

$$\underbrace{\sum_i c_i \times G_i}_{\text{Generator Dispatch}} + \underbrace{\sum_j c_{I_j} \times I_j}_{\text{Interconnector Dispatch}} \leq \underbrace{\mathbf{LR}_{\text{N-1},m}}_{=\sum_{n \in \text{N-1}} \text{LR}_{n,m}} - \underbrace{\varepsilon}_{\text{Operating Margin}}$$

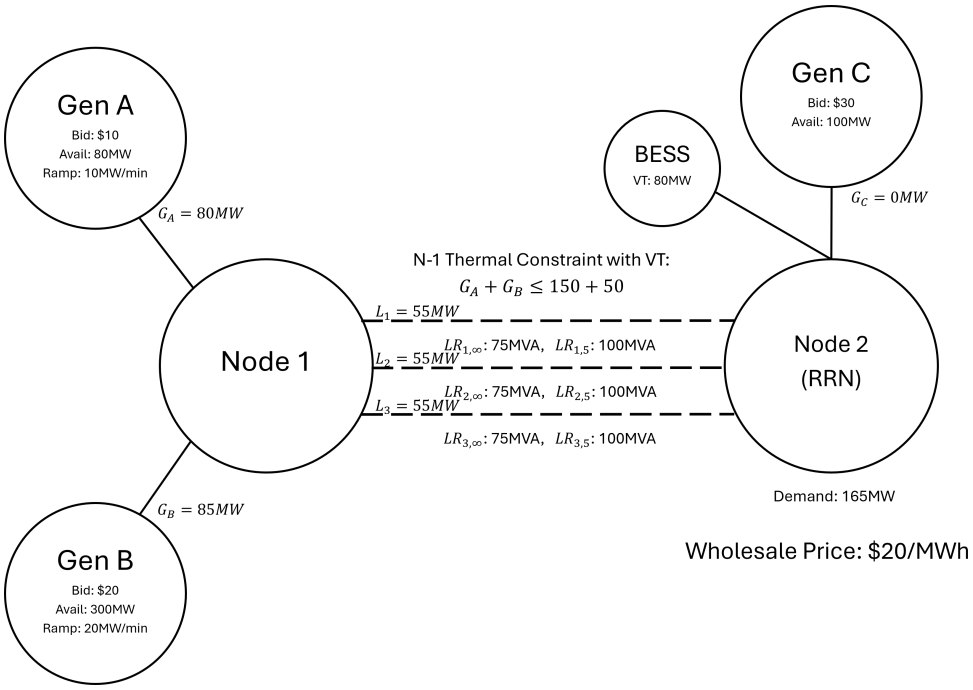
In this example, generator contribution factors to corridor flows (c_i) must be 1, while Panel A introduces continuous line ratings ($\text{LR}_{n,\infty}$) whereby each line can be operated safely at 75MVA in perpetuity. Ignoring the operating margin ($\varepsilon = 0$), the thermal constraint collapses to:

$$G_A + G_B \leq 150$$

We now find the cheapest market clearing and dispatch across generators to satisfy 165MW of demand at Node 2, equivalent to SCED in a broader setting. Generators A, B and C are in ascending cost order. We dispatch Generator A at its maximum 80MW availability and note that Generator B's 300MW availability is sufficient to clear demand, absent the thermal constraint. Given Generators A and B are limited to 150MW, we dispatch Generator B at 70MW and C to fulfill the final 15MW. The



(a) NEM market clearing



(b) NEM market clearing with virtual transmission

Fig. 1: Virtual transmission worked example. **a**, market clearing without virtual transmission. SCED dispatches the cheapest combination of generators to satisfy demand subject to the N-1 thermal constraint. The solution dispatches 80MW of A, 70MW of B and 15MW of C. The wholesale price is $30/MWh. **b**, market clearing with virtual transmission. With a virtual transmission battery at Node 2, 50MW is added to the constraint and demand is met entirely by A and B.

market-clearing wholesale price is Generator C's bid of \$30/MWh.

In Panel B, we simulate a successive time interval with VT, allowing it to relax the thermal constraint by enabling a shorter duration m'-minute line rating ($\mathbf{LR}_{\text{N-1},m'} > \mathbf{LR}_{\text{N-1},m}$). A battery connected to Node 2 ("downstream" of the monitored lines) reserves V for an N-1 outage. When armed ($\mathbb{1}_{\text{VT}} = 1$), a potential m'-minute VT expansion ($\gamma_{m'}$) is a function of: the increase in maximum flow along the corridor (I), battery capacity reserved to respond to residual demand following an outage (II), and upstream generation which can be backed off within m' minutes (III). The system operator applies an additional operating margin (ε^{I}-ε^{III}), a hurdle which I-III need to clear to enable expansion.

$$\underbrace{\sum_i c_i \times G_i}_{\text{Generator Dispatch}} + \underbrace{\sum_j c_{I_j} \times I_j}_{\text{Interconnector Dispatch}} \leq$$

$$\underbrace{\mathbf{LR}_{\text{N-1},m}}_{m\text{-minute Corridor Limit}} + \underbrace{\gamma_{m'}}_{\text{VT Expansion Function}} \times \underbrace{\mathbb{1}_{\text{VT}}}_{\text{VT Switch}} - \underbrace{\varepsilon}_{\text{Operating Margin}}$$

$$\gamma_{m'} = \left[\min(\underbrace{\mathbf{LR}_{\text{N-1},m'} - \mathbf{LR}_{\text{N-1},m}}_{\text{(I)}} - \varepsilon^{\text{I}}, \underbrace{c_{\text{BESS}} \times \text{V}}_{\text{(II)}} - \varepsilon^{\text{II}}, \underbrace{\text{Recoverable Generation}}_{\text{(III)}} - \varepsilon^{\text{III}})\right]_+$$

Per Panel B, VT enables a 5-minute line rating ($\mathbf{LR}_{\text{N-1},m'} \Rightarrow \mathbf{LR}_{\text{N-1},5} = 200$), aligning with the NEM's 5-minute settlement. Generators A and B can each be respectively ramped down by 50MW and 100MW within five minutes. Recoverable Generation, therefore, is the summation across Generators A and B of their greatest possible reduction in power within five minutes from current levels ($G_A = 80, G_B = 70$). Again, ignoring any operating margin ($\varepsilon^{\text{I}}, \varepsilon^{\text{II}}, \varepsilon^{\text{III}} = 0$) and with a unit battery contribution factor ($c_{\text{BESS}} = 1$), our realized VT expansion equation collapses to:

$$\gamma_5 = \left[\min(\underbrace{200 - 150}_{\mathbf{LR}_{\text{N-1},5} - \mathbf{LR}_{\text{N-1},\infty}}, \underbrace{80}_{c_{\text{BESS}}\text{V}}, \min(\underbrace{80}_{G_A^{\text{current}}}, \underbrace{50}_{G_A^{\text{ramp}}}) + \min(\underbrace{70}_{G_B^{\text{current}}}, \underbrace{100}_{G_B^{\text{ramp}}}))\right]_+ = 50$$

With a 50MW expansion in the thermal constraint:

$$G_A + G_B \leq 200$$

Re-solving the market with unchanged demand, Generator A maintains output ($G_A = 80$) and Generator B ramps by 15MW ($G_B = 85, G_C = 0$). The thermal constraint is no-longer binding and the wholesale price of electricity is Generator B's \$20/MWh bid. Between intervals, VT has enabled a reduction in the wholesale spot price of energy by \$10/MWh. Therefore, we can calculate the reduction in settlement

payments from VT as the reduction in electricity cost to load: $\$10/\text{MWh} \times 165\text{MW} = \$1,650/\text{h}$.

A Tale of Two Projects

Our two case studies establish a precedent for a competitively tendered contract for non-contestable, fixed-term VT service delivery, absent any formal investment test. Both contracts procure a subset of the battery's total capacity, allowing it to trade the remaining and uncontracted capacity freely, such as for energy arbitrage in the wholesale market. In this section, we will demonstrate that the projects, thus far, have failed to generate *any* consumer price benefits (Lessons 2-3)[4]. Subsequently, we address prevailing questions on the suitability of non-contestable, fixed-term service delivery as opposed to a contestable market service (Lesson 4) and, separately, how regulatory investment tests for VT could reject projects that provide a region's consumers net benefits (Lesson 5).

Table 1 summarizes the "System Integrity Protection Scheme" (SIPS) contracts for VBB and WSB and ensuing VT services. These services explicitly partition a proportion of the battery's tradable dispatch capacity (MW) and stored energy (MWh) for VT services, with the remainder available for merchant operation. The procurement of these VT services, and their treatment as transmission, are of central relevance to our subsequent discussion. Both projects required the coordination of the state government, transmission network service provider (TNSP) and market operator; they were not subject to Australia's Regulatory Investment Test for Transmission (RIT-T), but instead followed from state government legislation[5]. VBB's service, aims to "allow additional import of electricity over the Victoria to New South Wales Interconnector (VNI) of up to 250MW at peak times between November and March," with WSB's service "allow[ing] transmission lines supplying the Hunter, Sydney and Illawarra regions to run at a greater capacity"[20],[21]. Both the VBB and WSB projects precede physical transmission projects currently in planning[6].

Modeling commissioned by the Victorian Government prior to VBB estimates \$190m[7] in electricity price consumer benefits, with an additional \$12m in consumer benefits from avoided unserved energy[24]. One month prior to commissioning, AEMO indicated that VBB would enable VNI to run on a 5-minute line rating rather than a typical 15-minute rating[25]. Having established context on project procurement, we now address questions of, A) *how much have these VT services expanded constraints?* and B) *how effective have these services been in lowering the cost of dispatch to load?*

[4] We do not quantify any benefits from avoided load-shedding or similar if these assets have provided a contingency response following an N-1 line outage.

[5] National Electricity (Victoria) Amendment Bill 2020 in Victoria, and Energy Infrastructure Investment Act 2020 (NSW).

[6] The VNI West interconnector expansion project is expected in late 2030[22], and the New England REZ Infrastructure Project is expected in 2034[23].

[7] Real 2020.

Table 1: Asset and virtual transmission (SIPS) contract information for the Victorian Big Battery and the Waratah Super Battery.

	Victorian Big Battery	Waratah Super Battery
Commissioning	8 December 2021	Stage 1: 31 July 2025 (350MW) Stage 2: 8 June 2026 (700MW)
Region	VIC	NSW
Nameplate Capacity	300MW/450MWh	850MW/1680MWh
Connection Point	Moorabool[a]	Munmorah[b]
SIPS Commencement	29 November 2021	31 July 2025
SIPS Capacity	250MW, up to 30 minutes continuous dispatch	Intra-day and seasonally sculpted profile, up to 700MW and 2 hours continuous dispatch
Contract Tenure	1 Nov-31 Mar, 2021-2032	2024-2031
Contract Remuneration	\$162m total remuneration, paid over 11 years[c]	\$513m total nominal remuneration for WSB and Paired Generation, paid 2024-2031[19]
Additional Capital Costs	Not specified	\$105m total nominal transmission network augmentation costs, incurred 2024-2029
Recoverable Generation	VNI[d]	Metz Solar Farm (METZSF1), Sapphire Wind Farm (SAPHWF1)[e]
Constraints Expanded	Thermal constraints on the Victoria-NSW Interconnector (VNI)[f]	Thermal constraints in northern NSW/New England Renewable Energy Zone [g]

[a]See MLTS in Extended Data Figure 5.
[b]See Extended Data Figure 7.
[c]\$12.5m per period for 11 periods (real 2021). \$162m total contract value as at 2025 (indexation at 19.4% from Reserve Bank of Australia CPI).
[d]Recoverable Generation for VBB VT is not explicitly specified. However, VNI's simultaneous treatment as a line and dispatchable quantity within SCED allows for flow targeting down to a 5-minute granularity. When required, AEMO will directly impose constraints on VNI's target flow.
[e]See METZSF1 and SAPHWF1 in Extended Data Figure 7.
[f]See MURRAY-DDTS-SMTS in Extended Data Figure 6.
[g]See ARMIDALE-TAMWORTH-LIDDELL in Extended Data Figure 7.

Lesson 2: Contracted megawatts of virtual transmission ≠ megawatts of transmission expansion

Lesson 1 generalizes an m'-minute expansion as a function of three quantities, which is always weakly less than the battery's reserve capacity.

$$\gamma_{m'} = \left[\min(\mathbf{LR}_{\text{N-1},m'} - \mathbf{LR}_{\text{N-1},m} - \varepsilon^{\text{I}}, c_{\text{BESS}} \times \text{V} - \varepsilon^{\text{II}}, \text{Recoverable Generation} - \varepsilon^{\text{III}})\right]_+$$

With the AEMO Electricity Data Model database[26], we map the implementation of historical VT expansion to this framework from service commencement to 31 January 2026 (Table 2). We thereby observe the distribution of constraint expansion from VT, a more telling statistic on service efficacy than contracted battery capacity alone. Through this mapping exercise, we find that both VBB and WSB have almost unanimously failed to expand constraints in a combined 5 years of service, and consequently, neither has provided a measurable consumer benefit via constraint relief.

For the older asset, VBB's failing arises from use of the typical 15-minute line rating in expansion rather than the 5-minute rating, resulting in a zero megawatt expansion in all intervals ($\mathbf{LR}_{\text{N-1},m'} - \mathbf{LR}_{\text{N-1},m} = 0$). This has been the case throughout its 4.5 years of service. WSB's implementation is perhaps better characterized as mostly ineffective, given that it does enable expansion up to the 5-minute line rating on constraints and has generated positive expansion in 6 months of service. However, of the 4,400 hour study period, VT was only armed for a total of 6.25 hours in SCED constraints ($\mathbb{1}_{\text{VT}} = 1$). During this time, constraints were only expanded by a maximum of 30.6MW for 2.25 hours. During these intervals, the expansion of the constraints targeted by the VT service did not impact dispatch outcomes, suggesting that WSB has yet to generate a consumer price benefit. Aside from VT being scarcely armed, we also find that the operating margin ($\varepsilon^{\text{I}}, \varepsilon^{\text{II}}$) is often of a magnitude that eliminates constraint expansion (Figure 2 Panel A). This margin ensures that post-contingency headroom is preserved to maintain these corridors at their continuous limit when VT is armed[8]. Therefore, constraint expansion and the possibility for lower cost dispatch in the future appear to depend on whether it will be possible to relax this operating margin (Figure 2b).

Lesson 3: Virtual transmission benefits will only accrue at genuine transmission bottlenecks

Lesson 2 raises concerns on service productivity from issues in implementation. However, resolving these issues such that final expansion ($\gamma_{m'}$) is non-zero can only enable cheaper dispatch and consumer benefits when these constraints are otherwise binding. We find that WSB's service targets a material transmission bottleneck in northern NSW, whereas VBB's service targets constraints that typically do not limit VNI transfer capacity, as they only bound collectively for 6 hours across 4.5 years of service. We therefore conclude that even if the VT service provided by VBB were

[8]We discuss reserve deliverability and transmission headroom requirements in Extended Data.

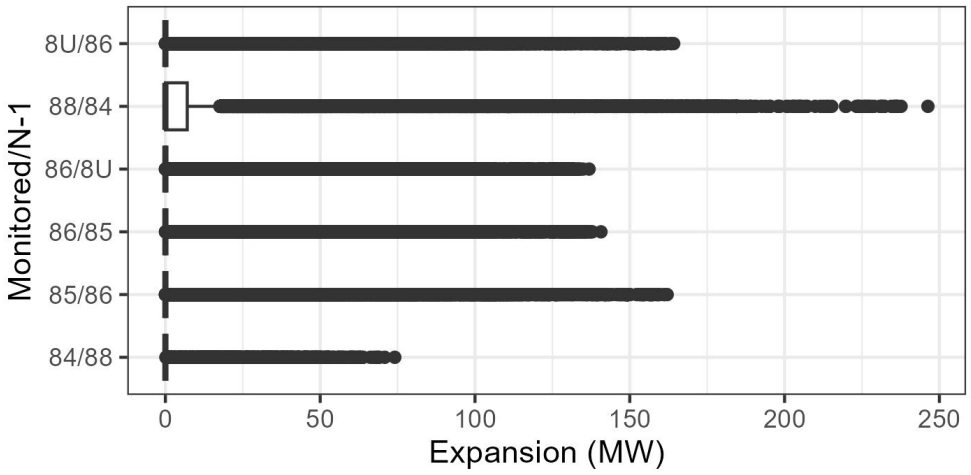
Table 2: Observed parameter values in VBB and WSB SIPS constraint implementation. Range given is minimum and maximum. VBB's expansion is ineffective, with expansion failing to utilize a higher line rating. WSB's implementation, by contrast, allows expansion up to a maximum 5-minute line rating. For WSB constraints, AEMO adopts a seasonal, day/night rating for 15-minute limits and a fixed rating for 5-minute limits. We find instances when the time-varying 15-minute value exceeds the 5-minute value.

	Victorian Big Battery	Waratah Super Battery
No. constraint equations	4	6
Constraint: Monitored/N-1[a]	• C1: DDMS1/DDMS2 • C2: DDMS2/DDMS1 • C3[b]: DDSM1/DDSM2, DDSM2/DDSM1 • C4[b]: EPTT/DDSM1, EPTT/DDSM2	• C1: 85/86 • C2: 86/85 • C3: 8U/86 • C4: 86/8U • C5: 84/88 • C6: 88/84
Default line rating duration	15-minute	15-minute
VT line rating duration	15-minute	5-minute
Terms		
(**I**) Line rating expansion	$\mathbf{LR}_{\text{N-1},15,t} - \mathbf{LR}_{\text{N-1},15,t} = 0$	$\mathbf{LR}_{\text{N-1},5,t} - \mathbf{LR}_{\text{N-1},15,t}$
(**II**) Battery reserve	$c_{\text{VBB}} \times V$	$c_{\text{WSB}} \times V_t$
(**III**) Recoverable generation	–	$c_{\text{METZSF1}} \times G_{\text{METZSF1},t-1} + c_{\text{SAPHWF1}} \times G_{\text{SAPHWF1},t-1}$
VT operating margin	$\varepsilon^{\text{I}} = 0$, $\varepsilon^{\text{II}} = 20$	$\varepsilon_t^{\text{I}} = 0$, $\varepsilon_t^{\text{II}}, \varepsilon_t^{\text{III}} = \mathbf{LR}_{\text{N-1},15,t} - \mathbf{LR}_{\text{N-1},\infty,t}$
Expansion function	$\gamma_{15,t} = \left\lceil \min(0, c_{\text{VBB}} \times V - 20) \right\rceil_+ = 0$	$\gamma_{5,t} = \left\lceil \min(\mathbf{LR}_{\text{N-1},5,t}, c_{\text{WSB}} V_t + \mathbf{LR}_{\text{N-1},\infty,t}, c_{\text{METZSF1}} G_{\text{METZSF1},t-1} + c_{\text{SAPHWF1}} G_{\text{SAPHWF1},t-1} + \mathbf{LR}_{\text{N-1},\infty,t}) - \mathbf{LR}_{\text{N-1},15,t} \right\rceil_+$
Observed Values		
(**I**) Line rating expansion	C1–C4: [0, 0]	• C1 & C3: [−68, 577] • C2 & C4: [−125, 313] • C5: [−14, 561] • C6: [273, 561]
(**II**) Battery reserve	$V = 250$[c]	$V_t = [150, 350]$, intra-day profile[c]
(**III**) Recoverable generation	–	$G_{\text{METZSF1},t-1} + G_{\text{SAPHWF1},t-1} = [0, 363]$
VT operating margin	$\varepsilon^{\text{I}}, \varepsilon^{\text{II}} = 20$	$\varepsilon_t^{\text{II}}, \varepsilon_t^{\text{III}}$: • C1 & C3: [104, 315] • C2 & C4: [127, 306] • C5: [162, 425] • C6: [0, 384]
Expansion	$\gamma_{15,t} = 0$	$\gamma_{5,t} = [0, 30.6]$

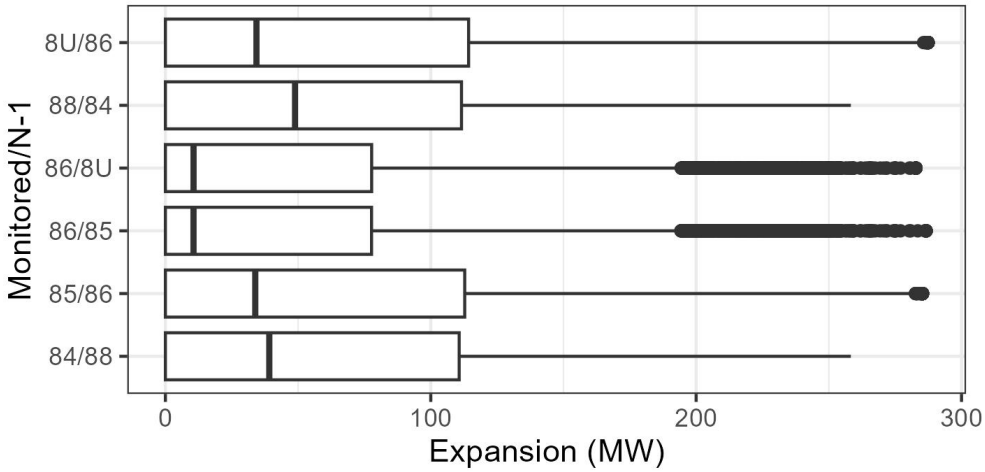
[a]See Extended Data Figures 6 and 7.

[b]Constraint takes the more restrictive Monitored/N-1 pair.

[c]Capacity when in service during asset VT period.



(a) Potential WSB VT expansion as implemented ($\gamma_{5,t}$)



(b) Potential WSB VT expansion ($\gamma_{5,t}$) without operating margin ($\varepsilon^{\text{II}}, \varepsilon^{\text{III}} = 0$)

Fig. 2: Potential WSB VT expansion by constraint with (**a**) and without (**b**) the VT operating margin ($\varepsilon^{\text{II}}, \varepsilon^{\text{III}} = 0$), unconditional on arming status ($\mathbb{1}_{\text{VT}}$). Boxes span interquartile range (IQR), whiskers extend to the furthest point from $1.5 \times$ IQR of the box. **a**, the presence of $\varepsilon^{\text{II}}, \varepsilon^{\text{III}}$ results in a non-positive expansion for constraints most of the time. **b**, removing the operating margin allows expansion above 100MW in roughly 25% of intervals.

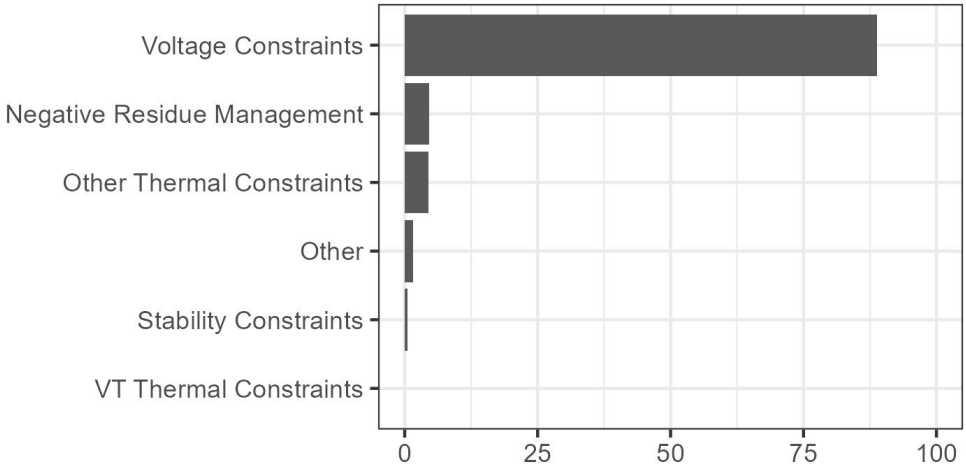
implemented in a way that expanded the constraints highlighted in Lesson 2, it was unlikely to return significant consumer benefits.

Given policymakers' intention of expanding VNI import capacity through VBB, we investigate constraint bottlenecks on VNI. During the 4.5 year VT period, VNI imports were actively restricted in 11% of intervals, mostly through **voltage stability** constraints. In Figure 3b, this occurs directly from binding **voltage stability** constraints in roughly 40% and indirectly in some 35% of cases through negative residue management constraints, which are imposed following excessive counter-price flows[9], typically from **voltage** constraints. VBB's VT service targets VNI **thermal** constraints, which were only binding for a total of 6 hours. In contrast, two pre-contingent **voltage stability** constraints bound for 851 hours collectively[10].

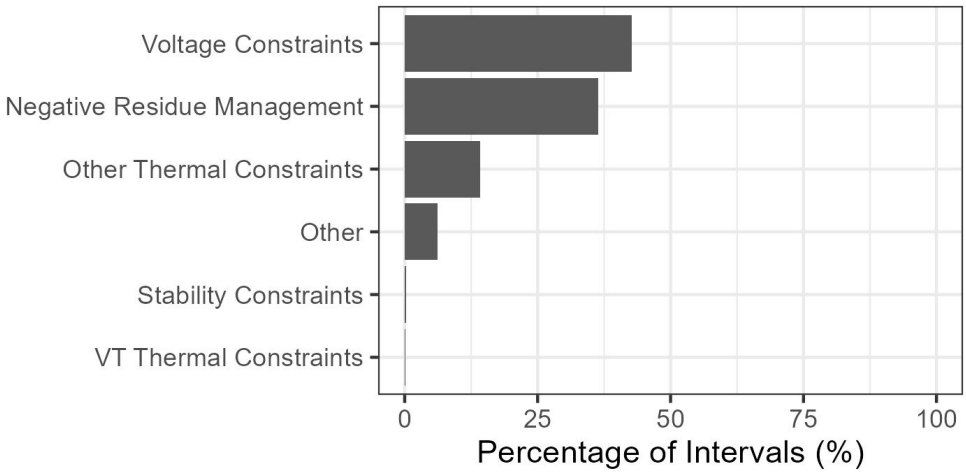
Reviewing the binding frequency of WSB constraints, we observe C1 and C2 binding for up to 86 hours each in the first 6 months of service (Extended Data Figure 8). This indicates that under more permissive implementation, WSB's VT service had potential to reduce these congestion costs and generate consumer price benefits. We relax WSB's VT operating margin and model a service with more permissive implementation in the following section.

[9] When an interconnector flows against prices, importing to a cheaper region from a more expensive region. Accumulated negative residues are recovered from the ratepayer of the importing region.

[10] Voltage collapse in southern NSW for loss of Basslink/Stockyard Hill WF (`N^^V_NIL_1` group), and voltage collapse in southern NSW for loss of Ararat to Ballarat lines (`N^^V_NIL_ARWBBA` group).



(a) VNI import limit by constraint type



(b) VNI import limit by constraint type, when import restricted

Fig. 3: VNI's maximum import in any given interval is set by network constraints. **a**, import limit by constraint type. Limits to VNI imports into Victoria are almost always set by a voltage constraint. **b**, import limit when VNI imports are restricted. Voltage constraints, directly or through negative residue management constraints, most often restrict imports. Thermal constraints outside of VBB's VT service have been limiting in 14% of intervals.

Counterfactual Virtual Transmission Scenarios

Lessons 2 and 3 describe why VBB and WSB VT services have failed to deliver a consumer benefit via the intended constraint-relief mechanism. These services have likely "left value on the table", and we use these lessons to construct a set of historical counterfactual scenarios for each service. To do this, we isolate historical intervals when a constraint was binding and re-dispatch the market with an expanded constraint within NEMPY[12] - a purpose-built SCED dispatch engine clone replicating the dispatch engine (NEMDE) used in practice. From this, we estimate a consumer benefit defined as the reduction in settlement payments incurred by load, and report regional and NEM-wide statistics. These exercises do not incorporate adjustments to bidding strategies, but are nonetheless instructive, returning an all-else-unchanged estimate of the historical value of specified constraint limit expansions.

We find that WSB "fails" Lesson 2 precisely because implementation is too restrictive to permit constraint expansion in most instances, but "passes" Lesson 3 as it does target a material network bottleneck. We model two scenarios for WSB with more lenient implementation and evaluate the potential consumer benefit from relaxation of the existing corridor. Scenario **W.1** removes the VT operating margin ($\varepsilon^{\mathrm{II}}, \varepsilon^{\mathrm{III}} = 0$) and arms expansion ($\mathbb{1}_{\mathrm{VT}} = 1$) whenever there is the possibility of expansion (Figure 2). However, during the time of study, WSB was unavailable for 44 days, owing to Stage 2 commissioning and a site fire, which coincided with high NSW wholesale prices and binding service constraints. Scenario **W.2** therefore extends **W.1** by simulating an environment where WSB is available and providing 350MW of service across the entire 183-day period.

VBB "fails" both Lessons 2 and 3. For completeness, we first allow the VBB service to expand currently targeted VNI **thermal** constraints by 250MW (**V.1**). This scenario simulates improved implementation, estimating a consumer benefit from the 6 hours when constraints did bind[11]. **V.2** and **V.3** target two different groups of frequently binding constraints that historically have limited VNI imports. Figure 3b shows that "other thermal constraints" (**thermal** constraints not currently targeted by VBB's VT service) have non-trivially restricted VNI imports, which incidentally relate to lines connecting directly to VBB's connection point[12], and at considerable distance from VNI[13] (Extended Data Figure 5). Therefore, **V.2** relaxes this constraint and two nearby, frequently binding and upstream **thermal** constraints[14]. **V.3** departs from **thermal** constraints altogether and simulates a VT service for the two pre-contingent **voltage stability** constraints that most commonly restricted VNI imports during our study window. It is a matter of other research to determine whether a VT service for these constraints is technically feasible; however, we seek to demonstrate the potential benefit from relieving these frequently binding

[11] This is the scenario described in the pre-implementation business case, which estimated $190m of consumer price benefits[24].

[12] Overloading on Elaine to Moorabool for loss of Ballarat to Moorabool, `V>>NIL_ELML_BAML2`.

[13] We discuss how bidding incentives under regional pricing can worsen network congestion and lead SCED to clear inefficient interconnector quantities in the Extended Data.

[14] Ararat to Waubra to Ballarat for loss of Kerang to Bendigo.

and high-value constraints[15]. In the absence of 5-minute line ratings and constraint contribution factors (c_{VBB}) for VBB, we assume constraints expand by 150MW in **V.2** and 100MW in **V.3** per 250MW of VBB capacity.

Table 3 lists counterfactual modeling results; regional and NEM-wide consumer benefits. Benefits are calculated as the change in settlement payments by load between the scenario and the existing *Base Case*, assuming bids remain unchanged. The *Base Case* entails the real-world implementation of all constraints. We independently run the *Base Case* through our model, to verify that our *Base Case* returns outcomes that coincide with real-world historical outcomes in Methods Table 5. We also report VT contract remuneration to calculate benefit-cost ratios (BCR), though we only observe VBB remuneration and not any other project or capital costs. Among the VBB scenarios, only **V.3** provides a large enough regional benefit to exceed contract payments to date (1.12 BCR). The results of **V.1** all but confirm poor value to the consumer, whereas **V.2** does lead to some $15m cheaper dispatch across 620 hours for Victorian consumers, but is well short of break-even for the ratepayer (0.23 BCR). WSB scenarios, providing two separate bounds of the same service, highlight the immense volatility in benefits. **W.2** regional benefit is 6.4x higher from increased service availability. The lumpiness of benefits are illustrated in Figure 4, whereby 85% of **W.2**'s benefit is incurred within 2 hours[16].

Lesson 4: Market-based compensation and competitive entry are potentially feasible for virtual transmission

VT's implementation via SCED makes market-based service compensation possible. Our scenario modeling demonstrates where potential for substantive market benefits from constraint alleviation may exist, with the potential to incentivize private market entry by batteries providing VT, alongside energy and other ancillary services. Importantly, a coupling of the market benefit and compensation can avoid the mismatch between ex-ante service remuneration and consumer market benefits observed in the established precedent. Though policymakers chose to bypass standard regulatory investment tests (which seek to quantify value for the ratepayer), there is no guarantee these services end up providing ratepayer value even when satisfying investment tests. Our scenario modeling not only demonstrates that benefits are highly volatile and can accrue in a handful of intervals, but also that they are sensitive to the terms of service delivery, such as asset availability and the precise implementation of the updated constraints (vis-a-vis WSB's existing implementation, **W.1** and **W.2**). Therefore, procuring VT via an inflexible ahead-of-time contract can result in an inefficient use of the battery's capabilities.

[15] Constraints for voltage stability in southern NSW are pre-contingent on the trip of generation elements in Victoria - Stockyard Hill WF, Basslink and Ararat to Ballarat lines (see Table 7 for full constraint equations). To this extent, we posit that a VT service could reduce the effective size of these contingent elements, not dissimilar to how thermal constraints are expanded.

[16] 15 minutes of >$14,000/MWh NSW wholesale price intervals fall below $1,000/MWh in **W.2** when WSB was in a planned outage and VT was unavailable during 20-25 November 2025. This 15-minute period would have saved NSW load some $23m from 90-165MW of alleviation on `N>>NIL_85_86_S` (C1 in Table 2).

Table 3: Regional and NEM consumer benefits for the VBB and WSB VT services. **Benefits** are calculated as the reduction in settlement payments paid by load between the scenario and *Base Case* (Methods), summed across each period. Regional figures are for VIC in VBB panel and NSW in WSB panel. **Hours** column corresponds to the number of intervals in which the market was re-solved when a scenario constraint was binding and subsequently relaxed per the scenario's implementation. **Obj. value** is the reduction in objective function value between *Base Case* and scenario, summed across each interval. See Methods Table 5 for comparison between *Base Case* and *Actual* settlement payments, and Extended Data Table 7 for full scenario construction details.

	Scenario	Updated Implementation	Updated Constraint Corridor	Regional benefit to load ($m)	NEM benefit to load ($m)	Obj. value ($m)[a]	Hours
	Observed benefit to date			0	0	–	–
	V.1	Up to 250MW expansion[b]	–	1.7	1.3	0.0	6
	V.2	150MW expansion	N-1 thermal constraints for Elaine to Moorabool and Ararat to Ballarat	15.1	25.4	22.2	620
	V.3	100MW expansion	Pre-contingent voltage constraints for voltage stability in southern NSW	74.8	15.9	3.0	785
	Contract remuneration to date			*-67.0*			
	Observed benefit to date			0	0	–	–
WSB	W.1	Removed VT operating margin[c]	–	5.1	5.2	0.2	174
	W.2	Removed VT operating margin[c] and full WSB availability at 350MW[d]	–	32.7	33.9	1.3	174
	Contract remuneration to date			*-31.3*			

[a] Excludes intervals with changes to constraint violation from *Base Case*.
[b] Uses existing VBB constraint contribution factors, such that $\gamma = c_{\text{VBB}} \times 250$.
[c] $\varepsilon^{\text{II}}, \varepsilon^{\text{III}} = 0$
[d] $V_t = 350$

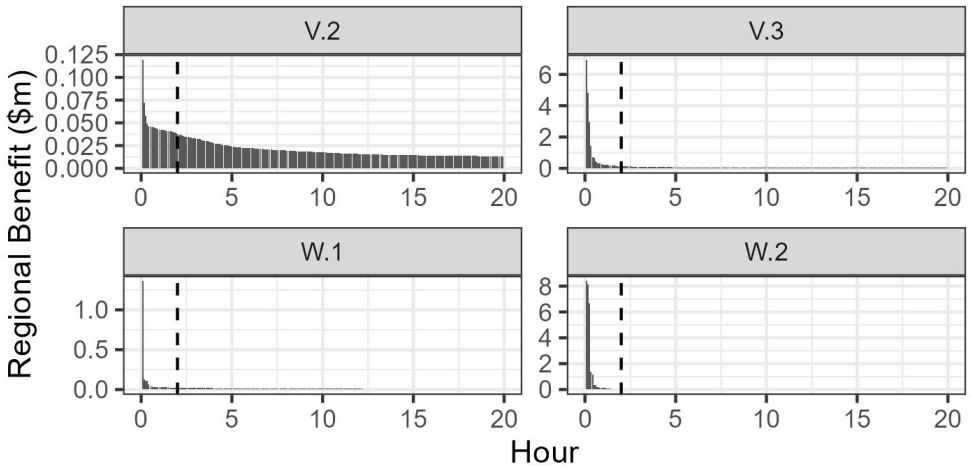


Fig. 4: Descending settlement payment reductions, by interval for V.2, V.3, W.1 and W.2 scenarios. We show only the top 20 hours (240 intervals) and mark the top two hours (dashed line). Benefits are highly skewed in all but V.2. One individual 5-minute interval accounts for \$0.1m of benefit in V.2, \$6.9m in V.3, \$1.4m in W.1 and \$8.4m in W.2. The top two hours (marked) in aggregate account for roughly 7.6% of benefit for V.2, 28.9% for V.3, 45.6% for W.1 and 85.2% for W.2.

A VT market would utilize SCED to co-optimize the value of additional energy throughput from the expansion of corridor limits against the as-bid cost of that expansion, in each interval. In the NEM, a conceptually similar precedent exists where energy procurement is co-optimized with ancillary market procurement volumes[27], and can result in reduced output of the largest generating unit where the marginal contingency reserve ancillary market costs exceed the marginal benefit from the largest unit's dispatch.

Integrating such a service within the existing regional pricing market design paradigm of the NEM will face challenges that we do not explore in depth in this paper. For example, because generators within a defined region face a common price, they cannot privately capture the constraint alleviation benefits that result from their actions, diminishing the incentives for batteries to locate in constrained parts of the network. However, establishing a VT market would necessarily entail pricing rules that depend on within-region location. It should be noted that introducing a VT market while maintaining a regional pricing market design would only allow the private capture of congestion relief benefits from expansion on the right side of constraints rather than equivalent relief through energy market actions on the left side of constraints. To this end, not all forms of constraint relief would be compensated equivalently.

Lesson 5: Costs and benefits from virtual transmission accrue unevenly across regions and market participants

SCED solves for the least as-bid system cost dispatch across the entire NEM, cross-subsidizing inter-regional supply and demand where it is efficient to do so, irrespective of the regional settlement outcomes that follow from the regional pricing rules. Similarly, a goal of transmission expansion, virtual or otherwise, is to improve system efficiency. This, however, can inherently create both cost-bearing and passive "winners", and "losers". We observe this clearly in **V.3**, estimated at $7.9m net benefits to Victorian load (cost-bearing "winner"), $23.1m and $8m to South Australia and Tasmania (passive "winners"), offset by losses of $55.8m and $34.3m in NSW and Queensland (passive "losers"). These transfers are coupled with an increase in median VNI imports by 81MW (Extended Data Figure 9).

However, having net benefits for the cost-bearer may not be a sufficient condition to satisfy social and market benefits tests for transmission, in many markets. The US lacks a central decision-maker to coordinate transmission, instead relying on regional transmission operators to reach consensus on interconnection. Interconnection has been historically motivated by reliability criteria, and despite FERC Order 1000 (2011) extending planning requirements to include market efficiency, Joskow (2022) argues this has motivated few (if any) projects[28]. Davis et al. (2023)[29] point to the difficulties between transmission operators to coordinate and agree on transmission value, precisely because projects create winners and losers. The issue of ratepayer value is further complicated in dual-use SATA, whereby ratepayers fund the entire project but receive energy market revenues[18]. Australia's standard regulatory process (RIT-T) takes a central view, enforcing a total surplus investment test that evaluates net benefits on a total-system basis, despite regional, consumer cost allocation. The issues from the perspective of a cost-paying ratepayer are twofold. First, a large subset of the consumer benefit, which is a transfer from producer profits toward avoided consumer settlement payments, is unaccounted for[17]. Second, regional costs are not allocated commensurate with benefits. This second issue came to a head with Marinus Link in 2023 when the Tasmanian Government reignited discussions on project ownership and cost allocation with Victorian and Federal Governments[18].

The costs, planning difficulties, and slowness of deploying transmission are well documented, as is the potential for substitution with storage (Denholm and Sioshansi, 2009; Davis et al., 2023)[30],[29]. We draw this comparison of transmission policy to highlight tensions in the procurement of transmission for market efficiency. Where a core component of traditional transmission's value stack has come from large displacements in generation capital costs[31],[32], infrequent transmission expansion under VT

[17] This is an artifact of regional pricing. In markets with nodal pricing rules, the difference in settlement payments made by load and collected by generators (reflecting the basis in nodal prices) is redistributed to the ratepayer as a merchandising surplus, or from auction revenues of the financial transmission rights that securitize this merchandising surplus.

[18] Marinus Link is an undersea interconnector project between Victoria and Tasmania. Similarly, the fate of a separate project, VNI West, seems to depend on upcoming 2026 Victorian state government elections rather than RIT-T cost evaluation and benefits estimation[4],[5].

is likely too small to provide these system planning benefits. The option value from deferred transmission lines could support VT in the RIT-T; both VBB and WSB service contracts precede major transmission projects. However, as stated in Lesson 4, including VT-providing batteries as regulated assets under rigid service delivery contracts forgoes the additional value from co-optimization with the battery's other market uses.

Policy Discussion & Conclusion

We review two world-first VT deployments, summarizing learnings across five lessons. Operationally, projects hold reserve capacity downstream of the transmission corridor, allowing for a conditional thermal limit expansion that enables higher transmission capacity within pre-contingent thermal constraints. We find that neither project effectively utilizes this mechanism, with both delivering poor value to the consumer via the intended constraint-relief mechanism. Our counterfactual modeling demonstrates the potential for both projects to have higher and even net positive consumer benefits. However, these benefits are volatile, and procurement VT services through rigid service delivery contracts results in a mismatch between benefits and ex-ante cost recovery. Further, rigid VT contracts can prevent the battery capacity from potentially providing more valuable energy services.

With most non-US grids around the world operating a regional pricing market design, VT could become an important tool for managing network congestion amid rising network constraints. However, if VT continues to be procured and funded as transmission (i.e. absent a market for short-run procurement), forecasting the impacts on dispatch is vital. The cases explored in this paper demonstrate the need for policymakers to apply regulatory processes that align compensation mechanisms to the expected (or delivered) benefits of VT if they are to deliver value to ratepayers.

Methods

Data Sources

Our research utilizes the AEMO Electricity Data Model 5.6.0 Market Management System[26] (MMS) and NEMWeb[33], which together collate historical NEM dispatch data, inputs and outputs. For the first section we scrape MMS tables for market prices (`DISPATCHPRICE`), constraint formulation (`GENERICCONSTRAINTRHS`, `SPDINTERCONNECTORCONSTRAINT`, `SPDCONNECTIONPOINTCONSTRAINT`), realized constraint terms and values (`NETWORKREALTIMERATING`, `NETWORKSTATICRATING`, `DISPATCHCONSTRAINT`), interconnector and generator values (`DISPATCHLOAD`, `DISPATCHINTERCONNECTORRES`), and from NEMWeb: analog network and generator telemetry, and SIPS term values. For the second section, we scrape dispatch inputs from MMS and NEMWeb through NEMPY[12] to re-dispatch the market under scenarios. Where we report historical actual dispatch objective values, this comes from the MMS `DISPATCHCASESOLUTION` table.

Market Re-dispatch Procedure

Scenarios require re-dispatch of the market through NEMPY[12], an open-source purpose-built NEM security-constrained economic dispatch environment in Python. The program constructs successive linear and mixed integer problems in sequence per AEMO's NEM dispatch procedure[34], which are solved with Gurobi 13[35].

The NEM's linear dispatch problem takes the following generic form.

$$\min_{G^b_{i,m}, I_j} \sum_i \sum_m \sum_b G^b_{i,m} \times \lambda^b_{i,m}$$

[objective function]

s.t.

$$G^b_{i,m} \leq \bar{G}_{i,m}$$

[unit capacity]

$$\sum_b G^b_{i,m} = G_{i,m} \leq \bar{G}_{i,m}$$

$$\sum_{i \in R} G_{i,\text{energy}} - \sum_{j \in R} (I_j + \mu^R_j I^{loss}_j) = D^R_{\text{energy}}$$

[energy balance]

$$\Delta G_{i,m} \leq \rho^u_{i,m}$$

[unit ramp]

$$\Delta G_{i,m} \geq -\rho^d_{i,m}$$

$$I_j = \sum_k w_k b_k$$

[interconnectors]

$$I_j^{loss} = \sum_k w_k f(b_k)$$

$$\sum_k w_k = 1$$

$$\sum_i G_{i,m'} = D_{m'} + \eta_{m'}$$

[ancillary service markets]

...

$$\sum_i c_i \Delta G_i + \sum_j c_{I_j} \Delta I_j \leq \mathrm{LR}_{l,15} - L_l - \sum_{l'} \delta_{l'} L_{l'} - \varepsilon_{(1)}$$

[network constraints]

...

This problem solves for system least cost for each given 5-minute interval (we drop the subscript t), by choosing the megawatt dispatch (G) for each unit i in each of i's 10 generation (and additionally, load) bid bands, b, across each of 11 (previously 9) wholesale and ancillary markets, m, and megawatt flow for each interconnector I_j. Objective function cost is defined as the sum of cleared loss-adjusted price bids ($G \times \lambda$) and (non-exhaustively) subject to a regional energy balance constraint, unit constraints, ancillary service constraints, interconnector and network constraints. We describe unit constraints for capacity, ensuring dispatched quantities do not exceed maximum capacities at the bid band ($\bar{G^b}$) and unit level ($\bar{G}$), and ramp rate constraints which bound the rate of change in unit-level dispatch. Ramp rates (ρ^u, ρ^d) are determined exogenously as a function of registered, unit-level values, bidded ramp rates and telemetered values. Energy balance constraints ensure that a region R's aggregate unit dispatch is sufficient to meet loss-inflated energy demand (D_{energy}), and the net sum of interconnector exports I_j inflated by R's share of losses, μI^{loss}. To preserve linearity, the program specifically selects interconnector flow (and loss) weights which form a linear combination of breakpoints, b_k ($f(b_k)$), for each I_j (I_j^{loss}). Ancillary services procurement occurs through a similar balance constraint. These markets are often procured system-wide and we omit formulations requiring regional ancillary service procurement (i.e. when a mainland region is electrically separated or Basslink is unable to transmit FCAS). Aggregate volumes procured, $\sum_i G_{i,m'}$, must equal the sum of a procurement target, D'_m, and adjustment term, $\eta_{m'}$, for each ancillary market m'. The procurement target is a constant (220MW for raise, 210MW for lower) for regulation markets and can be increased by an inter-temporal lagged function of time error[36]. For a subset of contingency markets, the procurement target usually varies around the largest individual generating unit. Ancillary services create additional unit ramp and capacity constraints due to the simultaneous provision of these services and energy which we omit here (see AEMO FCAS Model in NEMDE[37]). Likewise, for system constraints we describe an N-1 thermal constraint here in the same notation as previously. System constraints also include

linear constraint approximations for voltage stability, transient and oscillatory stability, and some state-dependent system strength and voltage constraints[38]. Given the large and sparse nature of the problem, the likelihood of infeasible solutions is non-trivial. AEMO prescribes constraint violation penalties which allow the solver to violate constraints (at cost) in order to find a solution.

Unlike most energy markets, the NEM has no scheduling market and as such, the dispatch procedure runs just prior to interval commencement. The procedure sequentially runs up to six permutations of this optimization problem[34]. A mixed integer program with non-convex start-up constraints for units which can start and synchronize rapidly determines the on/off status of these generators for a given interval, and a subsequent linear program with these on/off decisions allows dual-variable prices to be recovered. This procedure is run twice with Basslink HVDC frequency controller status per SCADA and switched off[34]. For each Basslink linear program solution, a re-run with relaxation of the violated constraint may be initiated if the violated constraint is an ancillary service market, intra-regional network or interconnector constraint, and a market price exceeds the market price cap or falls short of the market price floor[34].

Using this procedure, we re-dispatch the market for alternative scenarios by relaxing appropriate network constraints. For each scenario, we identify intervals in which a scenario constraint was binding and re-solve only those intervals. Other than the scenario's network constraints themselves (which are partially relaxed), we leave inputs otherwise unchanged including generator bids. Our analysis is therefore unable to capture changes in system cost and settlement payments from inter-temporal changes in bidding and dispatch that arise from constraint relaxation. Given that we are unable to reformulate bids and the heavy computational cost from re-solving the current set of scenario intervals, we consider these assumptions reasonable. We utilize ARDC Nectar Research Cloud virtual machines and re-solve scenario intervals in parallel.

Settlement Payments Calculation

To calculate settlement payments and recover consumer benefits against history, we first re-solve the market using historical inputs (*Base Case*) and demonstrate that model outputs track history. We note that our *Base Case* differs from a conceptually similar style of analysis that also uses a dispatch engine clone (for Italy), Graf et al. (2020)[39] – they assume participants engage in cost-reflective bidding to establish a competitive benchmark; we instead examine outcomes conditional on the observed bids of participants. The market prices that emerge are compared to the published Regional Original Price (ROP) field in AEMO MMS, after applying the relevant market ceiling and market floor caps. The ROP is otherwise equivalent to the Regional Reference Price (RRP) used in settlement calculation, unless the market is under administered pricing or an ex-post price adjustment has been made. We calculate a

region R's settlement payment in interval t with the following equation.

$$\text{settlement}_t^R = \frac{p_t^R}{12} \times (D_t^R - \sum_{i \in R} G_{i,t} \mathbf{1}_{G_{i,t}<0})$$

$$+ \sum_{R'} (I_{R' \to R,t} - I_{R' \to R,t}^{loss}) \mathbf{1}_{I_{R' \to R,t}>0} \times \frac{(p_t^{R'} - p_t^R)}{12}$$

Total regional load is given by adding D^R to dispatched scheduled loads, which are identified through negative dispatch targets, $G_i < 0$; all of who settle at the regional wholesale price, p^R. However, inflows from other regions, R', result in the cross subsidization of load in R (paying p^R) with generation in R' (receiving $p^{R'}$), and create a settlement residue between regions. We refer to $I_{R' \to R}$ as the net megawatt transfer from R' to R. Region R is the beneficiary of this settlement residue during a net inflow ($I_{R' \to R} > 0$) and when region R' has cheaper energy ($p^{R'} < p^R$). We ignore settlement from FCAS, including changes to FCAS market prices and cost recovery.

We, therefore, define a regional consumer benefit in interval t as the *Base Case* settlement payment *minus* the scenario settlement payment. Aggregating benefits across regions gives us the NEM consumer benefit. We report aggregated regional and NEM consumer benefits across broader period of scenario re-dispatch in Table 3.

Objective Function Value Calculation

The objective function value is equivalent to an as-bid system cost. However, generators' exertion of market power, 'disorderly bidding' and other factors decouples bids from their marginal cost. For example, renewable generators frequently bid down to the market floor (-$1,000/MWh) in order to prevent curtailment from binding network constraints. Similarly, coal plants will typically offer minimum generation levels required to stay on at the market floor. This frequently results in a negative as-bid 'system cost', decoupling from more appropriate measures of system cost which typically estimate spent fuel cost (and opportunity cost of energy for storage) or competitive benchmarks which assume marginal cost bidding. We choose to report this metric to signal the importance of constraint alleviation from an optimization perspective, distinct from a consumer benefits and settlement payments lens. As we observe in Table 3, scenarios with large reductions in settlement payments can have relatively smaller impact to the as-bid system cost, and vice versa.

In NEMPY, the objective function is unscaled and the objective function value is reported in $/hour. For interval t, an objective function value reduction is calculated as $\frac{\text{Base Case} - \text{Scenario}}{12}$, which converts the reduction into dollar terms. The per interval reduction is censored before aggregation in Table 3. We prescribe a censoring process due to the tendency of intervals with changes in total constraint violation penalties to dominate the aggregate objective function value by scenario. Intervals which satisfy any of the following criteria between the *Base Case* and the scenario are excluded from Table 3's objective value reduction column:

- Change in the constraint violation status of any individual constraint
- If a relaxed scenario constraint was in constraint violation in *Base Case*
- An interval incurred involuntary load shedding

We report both censored and uncensored aggregates in Extended Data.

Data Preparation

The research period spans 29 November 2021 to 31 March 2022, 1 November to 31 March from 2022-2025 and 31 July 2025 to 31 January 2026, spanning approximately 224,640 5-minute intervals – the SIPS contracts do not cover the full calendar year. We develop an exclusion criteria to handle intervals with corrupted or missing data, and market intervention. For AEMO 'intervention' intervals, we filter on the non-intervention (`INTERVENTION` = 0) dispatch solution and we remove intervals with market suspension (`MARKETSUSPENDEDFLAG` = 0). We do not make any administered pricing cap adjustments to modeled prices.

Intervention intervals span 59.75 hours of the research period, and 16.33 hours are removed from a market suspension flag. Administered pricing is present in 14.67 hours. We find 20.75 hours with missing or corrupted MMS data (2023-12-05 13:45 to 13:50, 2024-11-19 13:30 to 13:45, 2024-12-12 00:40, 2025-03-04 14:00, 2025-09-30 04:05 to 2025-01-01 00:10). After filtering on the superset of intervals in which a scenario constraint is binding, we are left with 1,556.25 hours (18,675 intervals) re-solved in the *Base Case*.

We also make small changes to NEMPY's package code to improve accuracy and robustness amidst broken and missing data. These changes are available in our clone of the NEMPY package repository.

Model Robustness

We compare the re-solved historical baseline to history to test model accuracy and robustness. Table 4 describes the distribution of wholesale price error by region ($p_t^R - \text{ROP}_t^R$), across the 1,556.25 hours of required re-dispatch. Approximately 50% of absolute price errors are under \$0.1/MWh, and under \$10/MWh in 90% of intervals across all mainland states. Model performance is weaker for Tasmania, which is frequency decoupled from the NEM, maintains its own frequency operating standards and must often procure its own ancillary services. Price outcomes for Tasmania are less important for this analysis and we do not investigate the weaker model performance. However, a possible culprit is NEMPY's hard-coded Basslink regional loss shares (which are not provided through MMS).

In Table 5, we compare the headline statistics for *Base Case* against actual; namely, regional and NEM settlement payments. For the superset of VBB intervals requiring market re-solve, error in aggregate settlement payments is within 0.5%. For WSB, the regional error is slightly higher at 2.3% from a \$2.6m settlement discrepancy.

Table 4: Distribution of historical-baseline price errors ($p^{\text{Base Case}} - p^{\text{ROP}}$, \$/MWh) by NEM region.

	QLD	NSW	VIC	SA	TAS
mean	0.03	0.50	-0.11	-0.41	-2.72
P1	-7.72	-12.6	-30.0	-28.3	-81.2
P5	-0.56	-1.96	-8.21	-5.71	-39.4
P25	0	0	-0.07	-0.01	0
median	0	0	0	0	0
P75	0	0	0	0	0
P95	0.33	2.95	8.06	2.59	12.2
P99	7.20	20	30.6	19.4	48.1

Table 5: Historical baseline validation. *Base Case* re-solves each service period with historical inputs and no change to network constraints; error is the deviation of *Base Case* from *Actual* AEMO MMS settlement. Both periods are restricted to the intervals re-solved for the counterfactual scenarios in Table 3.

Service period	Settlement	Actual (\$m)	Base Case (\$m)	Error (%)
VBB, 1,393 hours	VIC	689.8	686.6	−0.5
	NEM	3,045.6	3,035.4	−0.3
WSB, 174 hours	NSW	111.8	114.4	+2.3
	NEM	137.3	139.3	+1.5

Extended Data

Disorderly Bidding Incentives under Regional Pricing

In Lesson 3, we identify historically limiting constraints on VNI imports. We subsequently discuss a frequently binding thermal constraint (overloading on Elaine to Moorabool for a loss of Ballarat to Moorabool) which forms part of the **V.2** scenario. The constraint and affected network are at some distance to VNI despite setting imports for approximately 120 hours during the VBB service period. Reflecting the constraint's distance to VNI, VNI's coefficient (c_{VNI}) was a minor 0.08 until late 2024 when the term was dropped from the constraint. We believe that this warrants further discussion on how such a "minor" constraint can have limited imports as frequently as it did and more so than constraints with much larger VNI coefficients. We will demonstrate that this constraint's impact on VNI flows is an outcome of the "disorderly" bidding incentives, which itself is a feature of regional pricing[15]. In this instance, "disorderly" bidding refers to generator bids below their marginal cost (often at the -$1,000/MWh market price floor) when they face the risk of curtailment due to binding network constraints.

The left-hand-side (LHS) of overloading on Elaine to Moorabool for loss of Ballarat to Moorabool as at 2024-09-23 is described in Table 6, with interconnector terms at the top and then generator terms sorted by coefficient. We note that the majority of generators are either solar or wind farms, outside of the eight or so batteries.

When the constraint is at risk of binding, then Elaine, Mercer and Moorabool wind farms with a 1.0 contribution factor ($c_i = 1$) are at greatest risk of curtailment. This is because, to prevent constraint violation, SCED would only need to lower the dispatch target of these generators by 1MW for each unit reduction in the constraint LHS - compared to 12.5MW for VNI imports with a 0.08 coefficient ($12.5 = \frac{1}{0.08}$). SCED limits generators and interconnectors due to binding network constraints in the order of their volume-adjusted cost on the objective function from being turned down. For example, suppose our constraint is in violation by 1 unit, requiring *something* to be turned down in order to relieve the LHS by 1 unit. At a unit coefficient, turning down Mercer WF (located in Victoria) by 1MW reduces the objective function by Mercer's bid, $\$\lambda_{\text{Mercer}}$/h, but incurs the cost of re-dispatching 1MW elsewhere to satisfy load at Victoria's marginal price of electricity, $\$p^{\text{VIC}}$/h; the net increase to the objective function follows $\$(p^{\text{VIC}} - \lambda_{\text{Mercer}})$/h. The value of interconnectors is determined endogenously within dispatch; a 1MW decrease to VNI imports into Victoria saves $\$p^{\text{NSW}}$/h but incurs $\$p^{\text{VIC}}$/h. Thus, to achieve a unit reduction in the constraint LHS via VNI imports, SCED would incur $\$12.5 \times (p^{\text{VIC}} - p^{\text{NSW}})$/h, which is typically positive when VNI is flowing from a lower to a higher priced region ($p^{\text{NSW}} < p^{\text{VIC}}$).

All else equal, limiting the interconnector in favor of Mercer is more efficient when $12.5 \times (p^{\text{VIC}} - p^{\text{NSW}}) < p^{\text{VIC}} - \lambda_{\text{Mercer}}$. Let's consider a few cases. A) Mercer bids at the marginal price of electricity such that $p^{\text{VIC}} - \lambda_{\text{Mercer}} = 0$, then it is almost always preferred to curtail Mercer. B) Mercer bids at marginal cost (approx.

Table 6: Left-hand-side terms for constraint `V>>NIL_ELML_BAML2`.

Term	Factor
`VNI`	−0.080
`Murraylink`	−0.433
`ELAINWF1`	1.000
`MERCER01`	1.000
`MOORAWF1`	1.000
`BALB1`	0.653
`YENDWF1`	0.652
`ARWF1`	0.648
`CROWLWF1`	0.643
`BULBES1`	0.633
`BULBESG1`	0.633
`BULGANA1`	0.633
`KIATAWF1`	0.596
`MUWAWF1`	0.571
`MUWAWF2`	0.571
`BRYB1WF1`	0.542
`BRYB2WF2`	0.542
`KIAMSF1`	0.523
`COHUNSF1`	0.449
`GANNB1`	0.449
`GANNSF1`	0.449
`BANN1`	0.437
`WEMENSF1`	0.437
`KARSF1`	0.433
`YATSF1`	0.433
`MRTLSWF1`	0.381
`OAKLAND1`	0.357
`SALTCRK1`	0.357
`BHB1`	0.338
`BROKENH1`	0.338
`STWF1`	0.338
`LIMOSF11`	0.240
`LIMOSF21`	0.240
`SUNRSF1`	0.240
`GIRGSF`	0.184
`NUMURSF1`	0.184
`WUNUSF1`	0.184
`MTGELWF1`	0.143
`GLENSF1`	0.136
`GLRWNSF1`	0.136
`WINTSF1`	0.136
`DARTM1`	0.077
`MCKAY1`	0.077
`WKIEWA1`	0.077
`WKIEWA2`	0.077
`BULBESL1`	−0.633

$0/MWh for merchant renewable assets), then the spread between prices must be sufficiently small to not curtail Mercer, $p^{\text{VIC}} - p^{\text{NSW}} < 0.08p^{\text{VIC}}$. C) Mercer bids down to the market price floor (-$1,000/MWh) to maximize its chance of being dispatched, $p^{\text{VIC}} - p^{\text{NSW}} < 0.08p^{\text{VIC}} + 80$. Under C), regional pricing rules ensure Mercer earns positive profits provided $p^{\text{VIC}} > 0$, occurring when it and other generators employing a similar strategy are not the marginal unit. Under C) and when $p^{\text{VIC}} > 0$ (i.e., Mercer's "gamble" has paid off), the inter-regional price spread must neccessarily clear $80/MWh ($p^{\text{VIC}} - p^{\text{NSW}} > 80$) for SCED to consider VNI imports more valuable than Mercer's distorted value of dispatch, which can result in suboptimal inter-regional transfers. These bidding strategies as a feature of the NEM's regional pricing and market outcomes are discussed extensively in Katzen and Leslie (2024)[15]; and Holmberg and Lazarczyk (2015)[40].

Other Sources of Transmission System Redundancy

Identification of WSB's VT operating margin (ε^{I} and ε^{II}) in Lesson 2 warrants a further discussion on purposes of additional transmission system redundancy beyond pre-contingent network constraints discussed herein.

When expansion is positive and armed through WSB's service, a constraint's new corridor thermal limit is:

$$\gamma_5 = \min(\mathbf{LR}_{\text{N-1,5}} - \mathbf{LR}_{\text{N-1,15}}, c_{\text{WSB}}V - \varepsilon^{\text{II}},$$
$$c_{\text{METZSF1}}G_{\text{METZSF1}} + c_{\text{SAPHWF1}}G_{\text{SAPHWF1}} - \varepsilon^{\text{III}})$$

$$\Rightarrow \gamma_5 + \mathbf{LR}_{\text{N-1,15}} = \min(\mathbf{LR}_{\text{N-1,5}}, c_{\text{WSB}}V + \mathbf{LR}_{\text{N-1,15}} - \varepsilon^{\text{II}},$$
$$c_{\text{METZSF1}}G_{\text{METZSF1}} + c_{\text{SAPHWF1}}G_{\text{SAPHWF1}} + \mathbf{LR}_{\text{N-1,15}} - \varepsilon^{\text{III}})$$

Which, after identification of the operating margin ($\varepsilon^{\text{II}}, \varepsilon^{\text{III}}$), becomes:

$$\gamma_5 + \mathbf{LR}_{\text{N-1,15}} = \min(\mathbf{LR}_{\text{N-1,5}}, c_{\text{WSB}}V + \mathbf{LR}_{\text{N-1,}\infty},$$
$$c_{\text{METZSF1}}G_{\text{METZSF1}} + c_{\text{SAPHWF1}}G_{\text{SAPHWF1}} + \mathbf{LR}_{\text{N-1,}\infty})$$

This effectively ensures the corridor remains at its continuous thermal limit ($\mathbf{LR}_{\text{N-1,}\infty}$) following a line outage and corrective action from the VT. As the corridor is counterfactually maintained at its 15-minute limit ($\mathbf{LR}_{\text{N-1,15}}$) when expansion is not armed and no corrective action is provided by VT,f we notice that additional redundancy is required when corrective action is provided by VT than through the system operator's standard re-balancing procedure alone. The purpose of this redundancy is not made clear to us.

Biggar and Hesamzadeh (2026)[41] remark on transmission redundancy for the *secure* delivery of post-contingency reserves in the context of VT. This ensures that

transmission elements are kept within their safe operating limits, not only following the trip of a network element (such as in N-1 thermal constraints), but also following the dispatch of contingency reserves. Network constraints for the system post-contingency therefore explicitly solve for reserve procurement that guarantees feasible reserve deliverability following activation. Despite substantial literature on this topic[42],[43], power system adoption of these measures is nascent. Though the NEM maintains an extensive set of pre-contingent network constraints covering thermal and stability limits (such as voltage stability); it lacks post-contingency reserve deliverability constraints, nor does it explicitly maintain an operating margin for the purposes of contingent reserve deliverability. Instead, constraint operating margins typically procure additional headroom for modeling approximation and error, dispatch errors, non-conformance and real-time balancing (i.e, FCAS regulation service)[44]. Though Biggar and Hesamzadeh (2026)[41] find that the efficacy of VT can be limited by required redundancy for contingency reserve deliverability, we do not find evidence of this being the reason for WSB's large operating margin.

In Lesson 4, we discuss the possibility of market-based procurement for VT - that is the co-optimization of N-1 thermal limit transmission redundancy with the as-bid procurement cost of VT reserves. A potential market reform enabling this service may also prudentially establish reserve deliverability margins or constraints, or other headroom requirements on the transmission system.

Voltage Stability Constraint Formulation

We introduce a general form of pre-contingent, N-1 **thermal** constraints used in SCED in Lesson 1 to discuss the current implementations of VT. However, our **V.3** counterfactual scenario simulates similar constraint expansion for pre-contingent, **voltage stability** constraints. Therefore, we introduce a general form of these **voltage stability** constraints here, which pre-contingently protect against voltage collapse in southern NSW following the trip of a generation element in Victoria. As we demonstrate in the general form, these **voltage stability** constraints are similar to N-1 **thermal** constraints, which pre-contingently ensure thermal limit adherence following the trip of a line. It is beyond the scope of this research to identify the engineering feasibility of constraint expansion for voltage stability. However, we suggest that a VT service need not interact with the provision of voltage stability itself in order to expand these constraints, but instead provide a reserve mechanism to cover residual demand from the pre-contingent element.

The general form of a **voltage stability** constraint ensures that a linear combination of local generator and interconnector dispatch is below some adjusted voltage stability limit, K_t^*.

$$\underbrace{\sum_i c_i G_{i,t} + \sum_j c_{I_j} I_{j,t}}_{\text{Generator and interconnector dispatch}} \leq \underbrace{\mathrm{K}_t^*}_{\text{Adjusted voltage stability limit}} - \underbrace{\varepsilon}_{\text{Operating margin}}$$

The adjusted voltage stability limit (K_t^*) is calculated for each interval, t, as a function of initial dispatch variables (i.e., at the end of $t-1$) and exogenous terms.

$$\mathrm{K}_t^* = \underbrace{\mathrm{K}}_{\text{Voltage stability constant}} - \underbrace{\kappa_\Delta \Delta_{t-1}}_{\text{Credible contingency}} - \underbrace{\kappa_\mathrm{D} \mathrm{D}_{t-1}}_{\text{Local demand}} + \underbrace{\sum_k \kappa_{N_k} N_{k,t-1}}_{\text{Capacitor banks and reactors}} + \underbrace{\sum_{s \subset i} \kappa_{G_s} \mathbf{1}_{G_{s,t-1}>0}}_{\text{Synchronous generator reactive power}}$$

The voltage stability constant (K; MW) is adjusted by the size of the credible contingency (Δ; MW), local demand (D_t; MW), contributions in reactive power from local synchronous generation ($\mathbf{1}_{G_s>0}$), and capacitor banks and reactors in the network (N_k; MVAr).

For **V.3** we implicitly assume that a VT battery holding reserve capacity on standby would be able to reduce the size of the credible contingency, Δ_{t-1}. We consider the relaxation of two voltage constraint groups in **V.3**:

- `N^^V_NIL_1`: voltage collapse in southern NSW for a loss of the larger of Basslink imports to Victoria and Stockyard Hill Wind Farm;
- `N^^V_NIL_ARWBBA`: voltage collapse in southern NSW for the loss of Ararat to Ballarat lines (partially separating generation in north-west Victoria from Melbourne).

Extended Data Figure 5 shows VBB's proximity to Melbourne in relation to all three of these contingencies, whereby it could potentially provide a pre-contingent reserve.

Scenario Construction

We provide a more technical summary of the scenario construction in Extended Data Table 7, including the MMS constraints we modify and the exact way in which these constraints are modified. We also identify a few related and anticipated transmission projects expected to provide relief to these constraints.

Table 7: Counterfactual scenarios: MMS constraints and technical implementation details.

	MMS constraints	Definition & context	Implementation
Victorian Big Battery			
V.1	V>>NIL_MSDD1_MSDD2_SIPS, V>>V_NIL_1A_SIPS, V>SW_MSDD1_MSDD2_SIPS, V>>NIL_MSDD2_MSDD1_SIPS, V>>V_NIL_1B_SIPS, V>SW_MSDD2_MSDD1_SIPS, V>>NIL_DDSM_DDSM_SIPS, V>>V_NIL_3_SIPS, V>>NIL_EPTT_DDSM_SIPS, V>>V_NIL_4_SIPS	Current service as intended and per the pre-implementation business case[24]. Up to 250 MW expansion on VNI thermal constraints.	$\gamma = c_{\text{VBB}} \times 250$ $\mathbb{1}_{\text{VT}} = 1$
V.2	V>>NIL_ELML_BAML2, V>>BDBU_ELML_BAML2, V>>BAML2_ELML_BAML1, V>>BAML1_ELML_BAML2, V>>V_NIL_18, V>>NIL_ARWB_KGBE, V>>BDBU_ARWB_KGBE, V>>NIL_WBBA_KGBE, V>>V_NIL_9, V>>BDBU_WBBA_KGBE	Expansion for two sets of 220 kV line N-1 thermal constraints: • Overloading Ararat to Waubra to Ballarat for loss of Kerang to Bendigo • Overloading Elaine to Moorabool for loss of Ballarat to Moorabool Related: Western Renewables Link (new 500 kV line between WBTS and SYTS), expected 2030[45, 46].	$\gamma = 150$ $\mathbb{1}_{\text{VT}} = 1$
V.3	N^^V_NIL_1, N^^V_BUDP_1, N^^V_CNYS_1, N^^V_CNLT_1, N^^V_LTYS_1, N^^V_DTSS_1, N^^V_CNCW_1, N^^V_MSUT_1, N^^V_2_6X_3J_1, N^^V_MLNK_1, N^^V_BURC_1, N^^V_CTMN_1, N^^V_2+5_1, N^^V_LTMS_1, N^^V_3L+11_1, N^^V_DPWG_1, N^^V_4+07_1, N^^V_BADP_1, N^^V_JNWG_63_X5_1, N^^V_LTWG_1, N^^V_LTUT_1, N^^V_SM_SCAP, N^^V_NIL_ARWBBA, N^^V_MLNK_ARWBBA	Expansion for two sets of pre-contingent voltage stability constraints: • Southern NSW voltage stability for loss of the larger of Stockyard Hill WF and Basslink • Southern NSW voltage stability for loss of Ararat to Waubra to Ballarat lines Related: VNI West (new 500 kV line from SW NSW to NW VIC), expected 2030; PEC[a]; Victorian System Strength Requirements[47].	$\gamma = 100$ $\mathrm{K}_t^* = \mathrm{K} - (\kappa_\Delta \Delta_{t-1} - \gamma) - \kappa_{\mathrm{D}} \mathrm{D}_{t-1} + \sum_k \kappa_{N_k} N_{k,t-1} + \sum_{s \subset i} \kappa_{G_s} \mathbf{1}_{G_{s,t-1}>0}$ $\mathbb{1}_{\text{VT}} = 1$
Waratah Super Battery			
W.1	N>>NIL_85_86_S, N>>NIL_86_85_S, DSNAP_N>>NIL_86_85_S, N>>NIL_86_8U_S, N>>NIL_88_84_S, N>>NIL_84_88_S, N>>NIL_8U_86_S	Current service with updated implementation: • Removed VT operating margin ($\varepsilon^{\text{II}}, \varepsilon^{\text{III}} = 0$) Related: New England REZ Infrastructure Project, expected 2034[23]	$\gamma_{5,t} = \min \left\{ \begin{array}{l} \mathbf{LR}_{\text{N-1},5,t} - \mathbf{LR}_{\text{N-1},15,t}, \\ c_{\text{WSB}} V_t, \\ c_{\text{SAPHWF1}} G_{\text{SAPHWF1},t-1} \\ + c_{\text{METZSF1}} G_{\text{METZSF1},t-1} \end{array} \right\}$ $\mathbb{1}_{\text{VT}} = 1$
W.2	N>>NIL_85_86_S, N>>NIL_86_85_S, DSNAP_N>>NIL_86_85_S, N>>NIL_86_8U_S, N>>NIL_88_84_S, N>>NIL_84_88_S, N>>NIL_8U_86_S	Current service with updated implementation: • Removed VT operating margin ($\varepsilon^{\text{II}}, \varepsilon^{\text{III}} = 0$) • Full WSB availability at 350 MW Related: New England REZ Infrastructure Project, expected 2034[23]	$\gamma_{5,t} = \min \left\{ \begin{array}{l} \mathbf{LR}_{\text{N-1},5,t} - \mathbf{LR}_{\text{N-1},15,t}, \\ c_{\text{WSB}} 350, \\ c_{\text{SAPHWF1}} G_{\text{SAPHWF1},t-1} \\ + c_{\text{METZSF1}} G_{\text{METZSF1},t-1} \end{array} \right\}$ $\mathbb{1}_{\text{VT}} = 1$

[a] Project EnergyConnect.

Network Diagrams

We include three reference figures on the transmission network topology for VNI, south-western Victoria (including VBB), and northern NSW (including WSB).

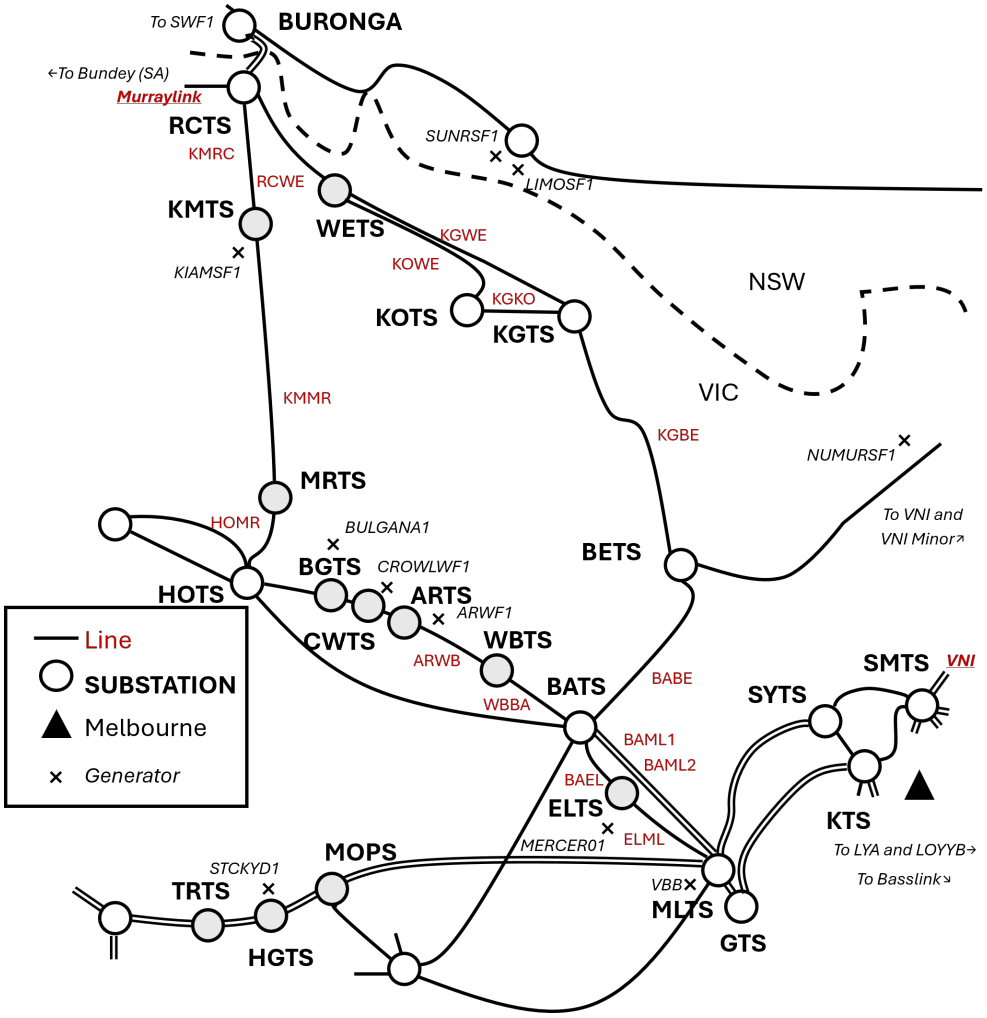


Fig. 5: Western VIC transmission network, running from the Heywood Interconnector (VIC-SA) in the west, Murraylink Interconnector (VIC-SA) and BURONGA (NSW) to RCTS (Red Cliffs in VIC) in the north-west into Melbourne. VBB scenarios explore constraint expansion for thermal constraints on ARTS (Ararat) to BATS (Ballarat), a vital transmission corridor into Melbourne, and voltage constraints on VNI (not pictured) in the east for the loss of ARTS to BATS, Stockyard Hill WF (STCKYD1) and Basslink (not pictured).

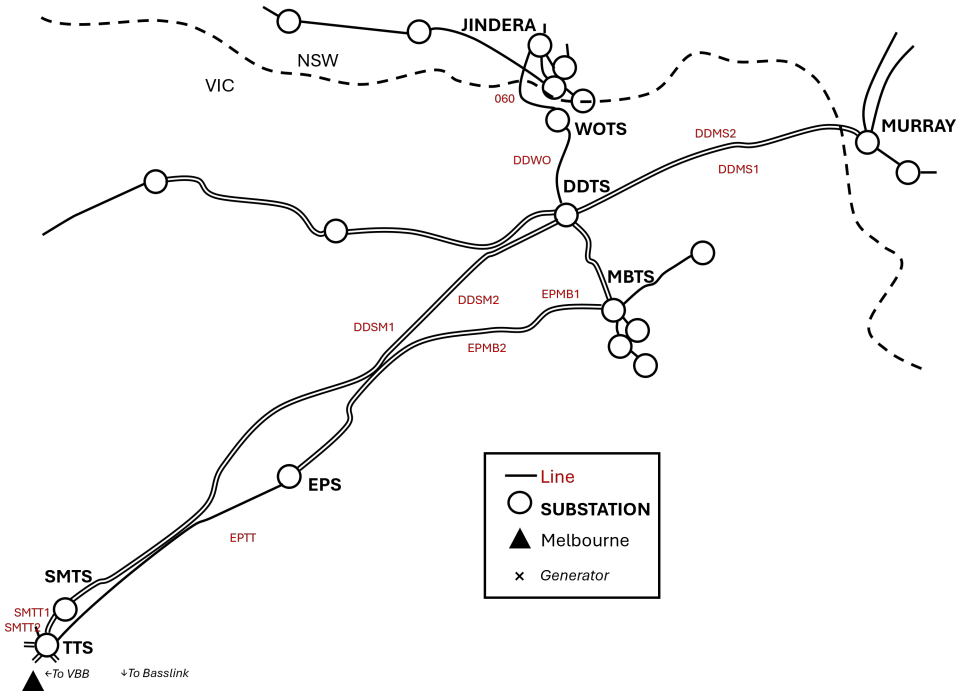


Fig. 6: VNI interconnector, running dual-circuit 330kV lines from MURRAY (NSW) to DDTS (Dederang in VIC) to SMTS (South Morang in VIC). Other pathways between NSW and VIC include VNI Minor, JINDERA (NSW) to WOTS (Wodonga in VIC) and Buronga to Red Cliffs (not displayed). VBB's virtual transmission targets thermal constraints on MURRAY-DDTS-SMTS that limit imports from NSW into VIC.

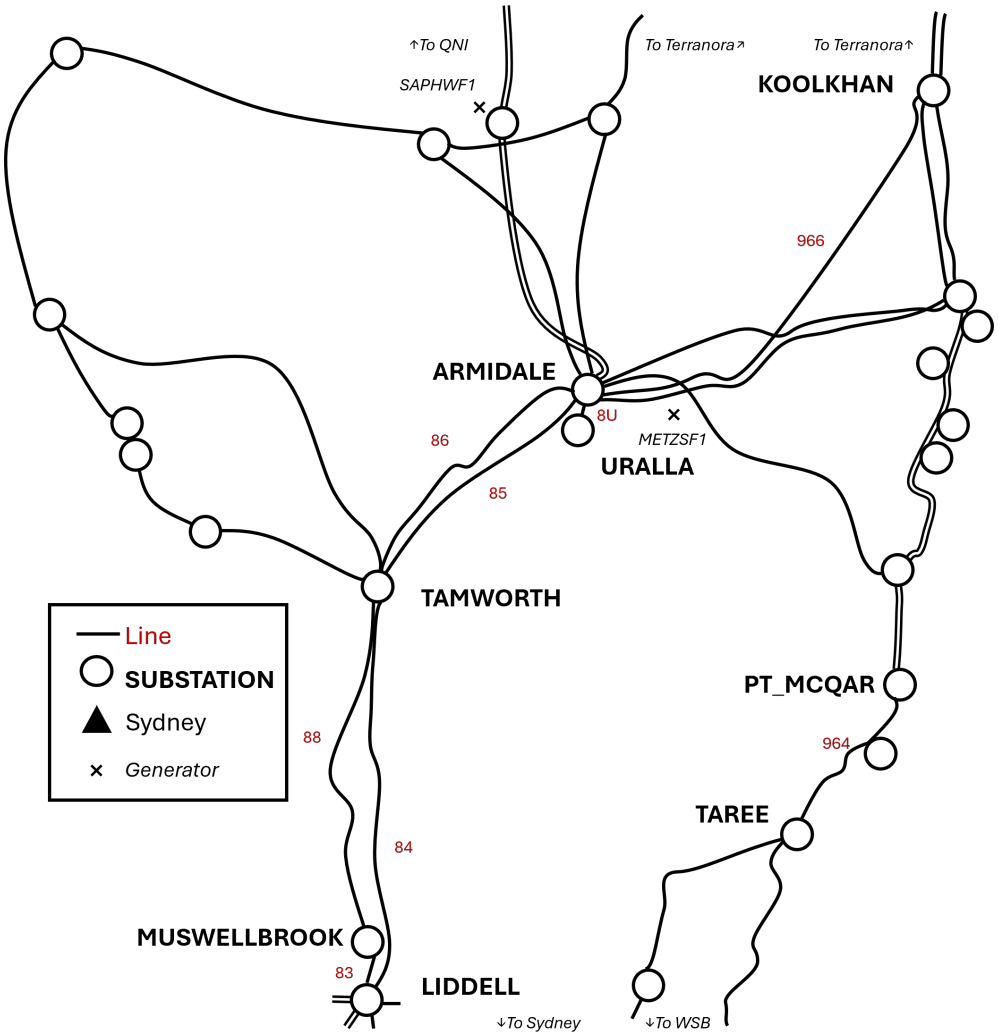


Fig. 7: Northern NSW transmission infrastructure, delivering power from Queensland imports along QNI to the north and surrounding renewable energy zone wind and solar assets. This 330kV corridor has frequently binding thermal constraints, limiting power transfer, which are relaxed through an upstream tripback scheme at Metz Solar Farm and Sapphire Wind Farm, and WSB located to the south-east.

Extended Data: Figures

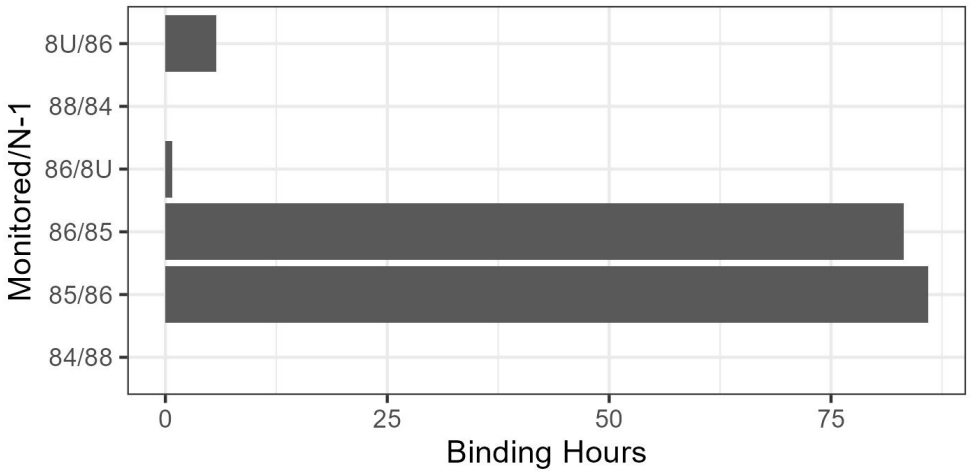


Fig. 8: WSB VT service constraint binding frequency. Binding hours by thermal constraint, 31 July 2025 to 1 February 2026. Parallel lines 85 and 86 (Armidale to Tamworth) binding frequently on a pre-contingent loss of the other.

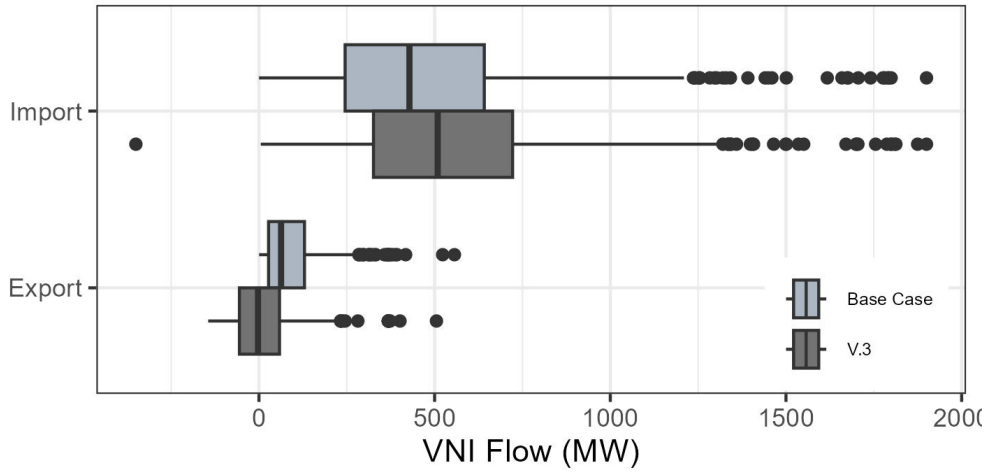


Fig. 9: Distribution of VNI flows, historically (*Base Case*) and **V.3** across 785 hours when **V.3** voltage constraints were historically binding. Median imports rise 81MW under **V.3** and the number of export intervals roughly halves.

Table 8: Counterfactual scenario settlement payment reductions by region ($m). Positive values denote a reduction in settlement payments relative to the *Base Case*.

	VBB			WSB	
Region	V.1	V.2	V.3	W.1	W.2
NSW	−0.45	4.26	−55.76	5.08	32.68
QLD	−0.12	1.81	−34.34	−1.17	−1.82
SA	0.09	2.37	23.12	0.05	0.12
TAS	0.01	1.89	8.05	0.11	0.18
VIC	1.75	15.12	74.85	1.15	2.73
Total	1.29	25.44	15.92	5.22	33.90

Table 9: Objective function value differences by counterfactual scenario ($m). Positive values denote an improvement (reduction) in the objective function value relative to the *Base Case*. "Total change" includes improvements from reductions in constraint violation penalties (CVP); "Excluding CVP changes" subtracts the contribution from changes in CVP.

	VBB			WSB	
	V.1	V.2	V.3	W.1	W.2
Total change	72.71	22.34	15.09	0.22	1.34
Excluding CVP changes	0.01	22.20	3.01	0.22	1.34

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Data and Code Availability
All data and models used in this analysis are accessible via Github. For scenario modeling, see https://github.com/loy-yang-c/Monash-NEMPY. For the NEMPY Python package with edits, see https://github.com/loy-yang-c/nempy-backend.